

Dominic Nguyen

Mechatronics Graduate

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PROJECTS

PLC Control Systems Project — UBC

SEP 2024 - NOV 2024

- Programmed ladder logic in ClickPLC to control a pneumatic piston system, achieving position accuracy within 2 mm of setpoint.
- Tuned PID coefficients via Ziegler-Nichols, manual, and autotune methods, analyzing tradeoffs between overshoot, rise time, and steady-state error.
- Built a SCADA interface in VTScada with real-time visualization and configured Modbus TCP/IP communication to third-party software.

Skills used: Ladder Logic, SCADA Design, PID Tuning, Modbus TCP/IP

3D Printed Biped Wheel Robot Pair Project — UBC

DEC 2025 - FEB 2026

- Designed and implemented a cascade PID control system, using Simulink to model and tune the inner and outer control loops before deployment.
- Coded firmware to estimate tilt angle using a Madgwick filter fusing accelerometer and gyroscope data to feed the control loop in real time.
- Applied multiprocessing on an ESP32 to run tilt estimation at 1 kHz and PID calculation at 500Hz, resolving I2C communication failures by switching to SPI to maintain stable balance for the full battery life.

Skills used: Simulink, C, 3D Printing, Sensor Fusion, RTOS, PID Tuning, Mechanical Design

PLA 25:1 Gear Ratio Cycloidal Reducer — *Personal Project*

JUL 2025 - AUG 2025

- Maximized gearbox efficiency to 89% by selecting low-friction materials and lubricating contact points.
- Identified a 0.24% skew error on the 3D printer axis and wrote a Python script to shift G-code instructions, eliminating the mechanical error.

Skills used: Mechanical Design, SOLIDWORKS, 3D Printing

VEX Gap-Crossing Robot, Hatch Hackathon — *UBC Senior Design Competition*

MAR 2026

- Re-engineered the drivetrain and switched to ultrasonic sensing to resolve recurring hardware failures under a 29-hour time limit, placing 2nd of 5 teams.

Skills used: Mechanical Design, Rapid Prototyping

LinkedIn:

[linkedin.com/in/dominicn17](https://www.linkedin.com/in/dominicn17)

Project Portfolio:

dominic817.github.io/portfolio/

EDUCATION

Graduated: May 2026

BASc. Mechatronics Engineering

University of British Columbia

Dean's List

Cumulative Grade: 80% (Dean's List)

SKILLS

CAD

SOLIDWORKS, Autodesk Inventor, Autodesk Fusion 360, Onshape

Manufacturing

3D Printing, Machine Tools, CNC Machine

Software

C, C#, MATLAB, Python, Windows Forms, Microsoft Excel

Instrumentation

Oscilloscope, Digital Multimeter, Circuit Analysis, Soldering

HOBBIES

Bouldering

Piano

Personal Projects

Video Games